HDOC-72: Attendance Management Program

Problem: Write a C program to calculate attendance percentages for multiple students and determine whether each student is eligible for examination (minimum 75% attendance).

Algorithm

1. Read the number of students.

2. For each student, read the total number of classes and classes attended.

3. Validate the input, then calculate attendance = (attended / total) x 100.

4. Print the percentage and examination eligibility based on the 75% threshold.

C Program

```
#include <stdio.h>

int main()
{
    int n, i, totalClasses, attendedClasses;
    float attendance;

    printf("Enter number of students: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        printf("\nStudent %d\n", i);
        printf("Enter total number of classes: ");
        scanf("%d", &totalClasses);
        printf("Enter number of classes attended: ");
        scanf("%d", &attendedClasses);

        if (totalClasses <= 0 || attendedClasses < 0 ||
            attendedClasses > totalClasses)
        {
            printf("Invalid attendance details.\n");
            continue;
        }

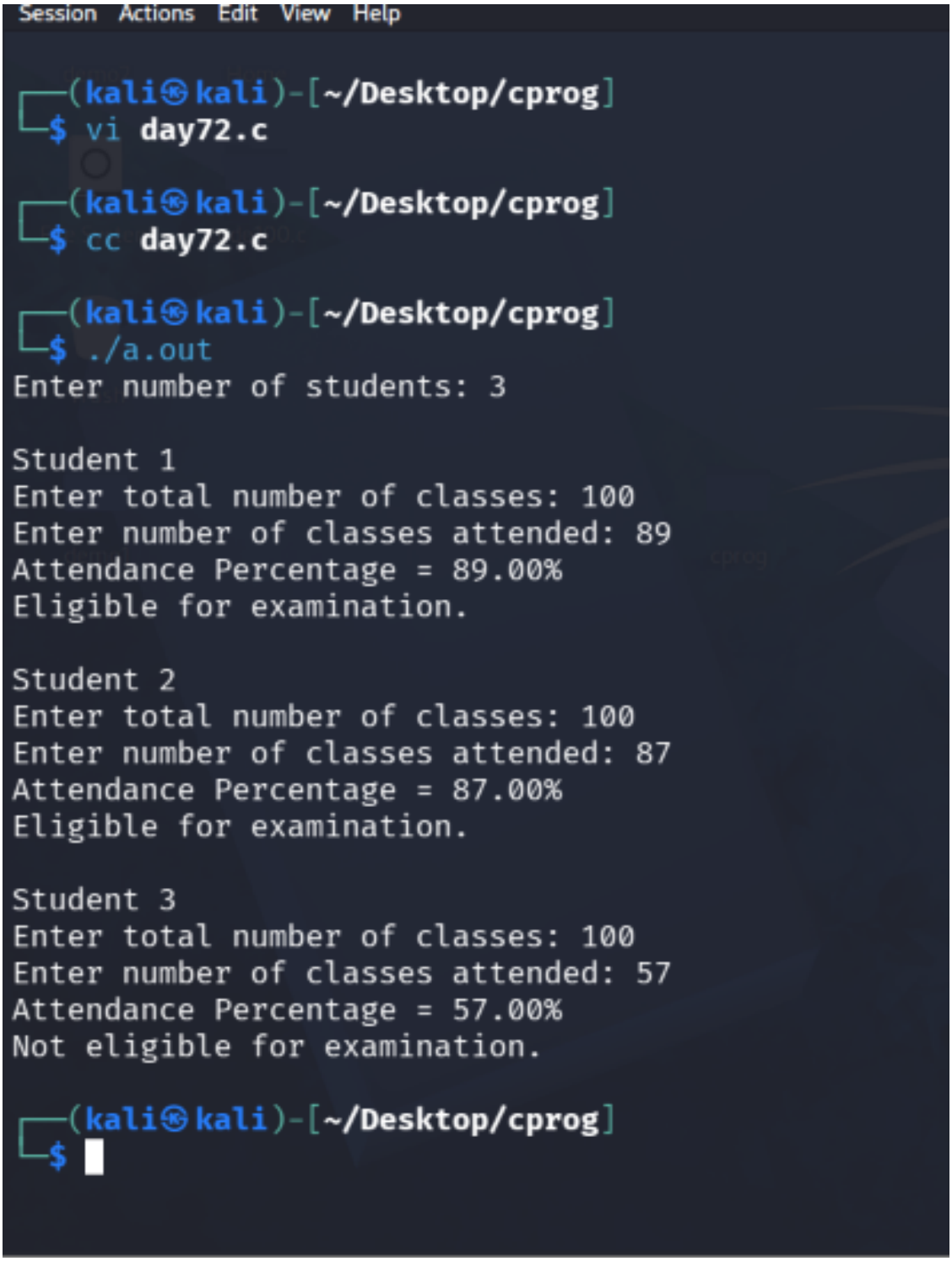
        attendance = (attendedClasses * 100.0) / totalClasses;
        printf("Attendance Percentage = %.2f%%\n", attendance);

        if (attendance >= 75)
            printf("Eligible for examination.\n");
        else
            printf("Not eligible for examination.\n");
    }
    return 0;
}
```

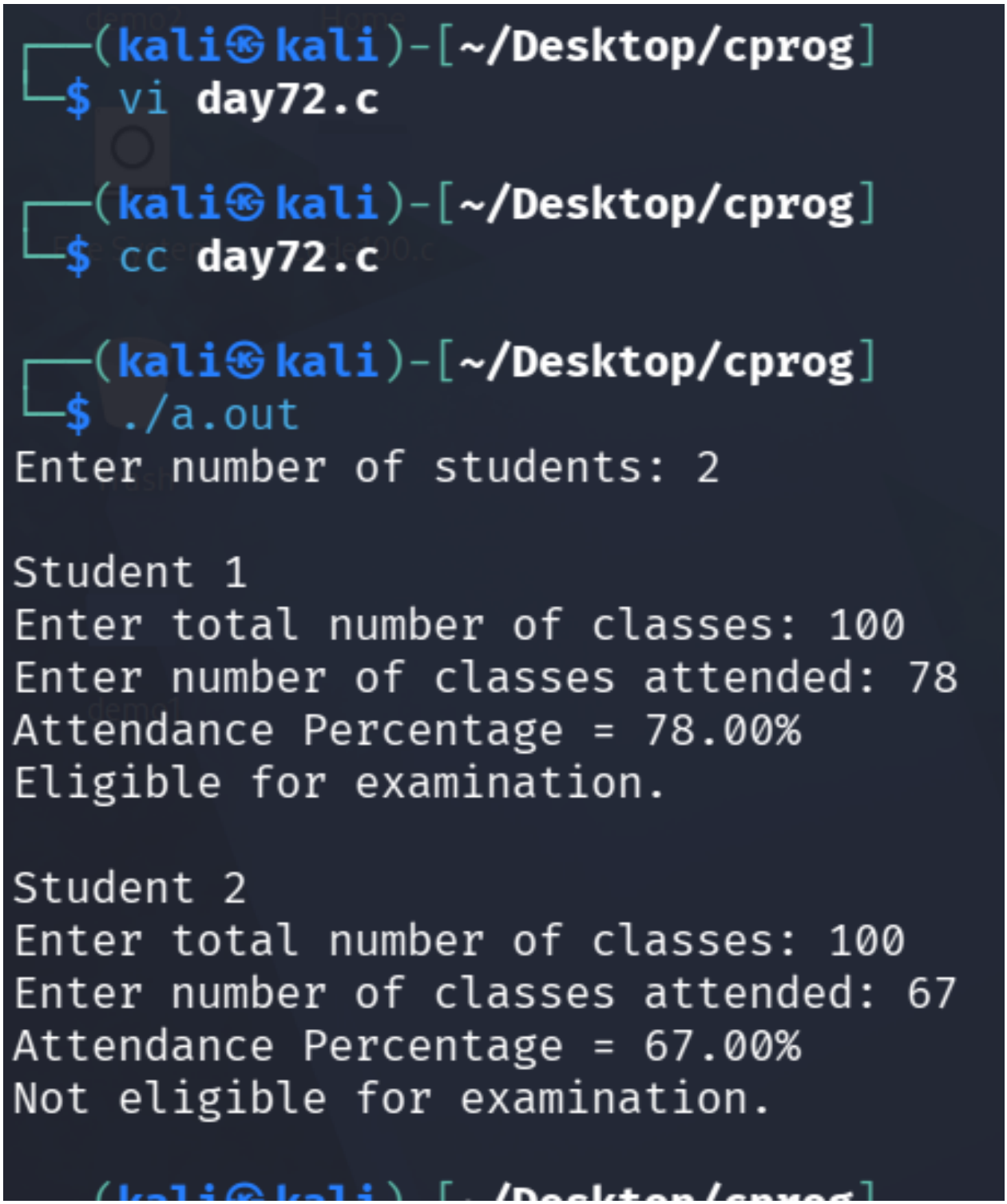
Test Case Table

Test case	Student	Total classes	Attended	Attendance	Result
1	1	100	89	89.00%	Eligible
1	2	100	87	87.00%	Eligible
1	3	100	57	57.00%	Not eligible
2	1	100	78	78.00%	Eligible
2	2	100	67	67.00%	Not eligible

Test Case 1 - Kali Terminal Screenshot


Session Actions Edit View Help
(kali㉿kali)-[~/Desktop/cprog]
$ vi day72.c
(kali㉿kali)-[~/Desktop/cprog]
$ cc day72.c
(kali㉿kali)-[~/Desktop/cprog]
$ ./a.out
Enter number of students: 3
Student 1
Enter total number of classes: 100
Enter number of classes attended: 89
Attendance Percentage = 89.00%
Eligible for examination.
Student 2
Enter total number of classes: 100
Enter number of classes attended: 87
Attendance Percentage = 87.00%
Eligible for examination.
Student 3
Enter total number of classes: 100
Enter number of classes attended: 57
Attendance Percentage = 57.00%
Not eligible for examination.
(kali㉿kali)-[~/Desktop/cprog]
$

Test Case 2 - Kali Terminal Screenshot



Result: The program calculates the attendance of each student and correctly reports examination eligibility using the 75% threshold.